SynaptOS

SynaptOS Architecture Proposal Pitch Deck

Retail Nervous System - Agentic AI that eliminates fresh-food waste and maximises perishable revenue autonomously.

MEMBER

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DATE

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COMPETITION

GDGoC 2026, Round 1: Ideation

The Vision Thesis

SynaptOS converts the greatest expense in the grocery industry the spoilage cost into the greatest revenue line with agentic AI that no one competitor or competitor around the globe has developed to date in the vocation of Southeast Asia.

$0.015
Cost per AI loop

333×
Min ROI per loop

8-16×
LTV:CAC ratio

17.2%
Market CAGR

The Problem

Fresh Food Is a Ticking Clock

Vietnamese grocery stores have 2-3% net margins with 15-25% of fresh produce wasted to spoilage. Current POS systems are simply paper registers with no intelligence of expiry clocks, demand variations or buying capacity.

15-25%

Structural Waste

15-25% of perishable GMV lost to spoilage annually

0

Passive Technology

POS systems record transactions but carry no pricing intelligence

500

Manual Decisions

Store managers can't monitor 500 SKUs in real-time

Why Vietnam, Why Now

The grocery market in Vietnam is projected to grow by 13.6% CAGR to about 1 trillion USD in 2033, the peculiarities of this rise require the act of spoilage to be considered acute:

Factor	Impact
Tropical climate (28-35°C)	30-40% higher spoilage vs temperate markets
Bach Hoa Xanh: 2,180+ stores	Scale multiplies problem across thousands of locations
E-invoice mandate (252K businesses)	Structured POS data creation at scale
Decision 222 (Circular Economy)	Enterprise demand for waste tracking is policy-driven

Core Architecture

1 SENSE

Real-time POS, weather, commodity prices, demographics

2 REASON

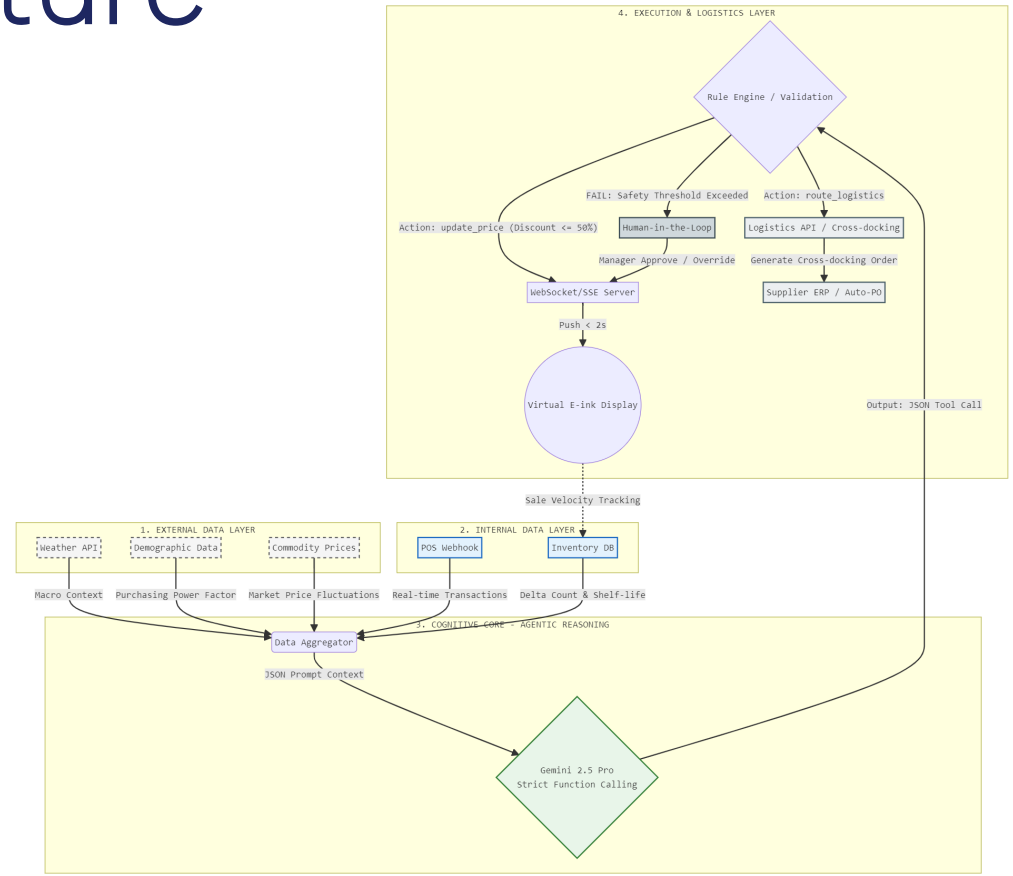
Calculate price elasticity, spoilage risk, optimal markdown

3 ACT

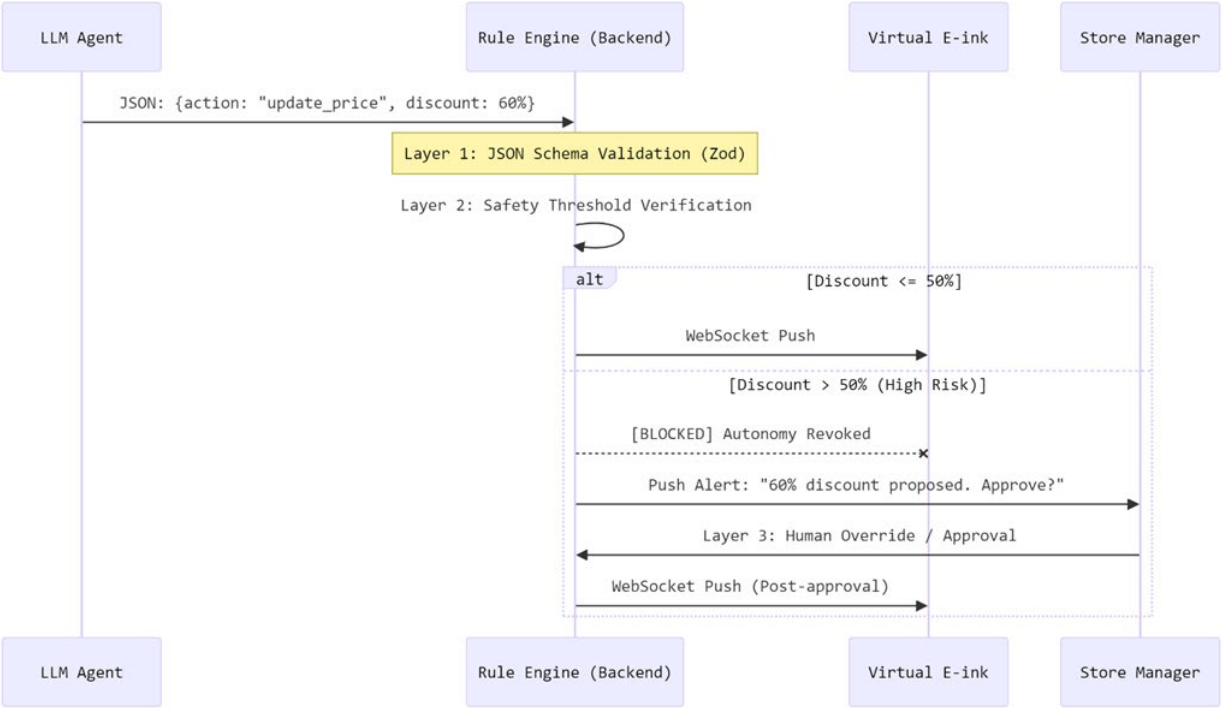
Push prices to E-ink labels, draft POs, trigger campaigns

4 LEARN

Measure velocity, self-calibrate, correct phantom inventory

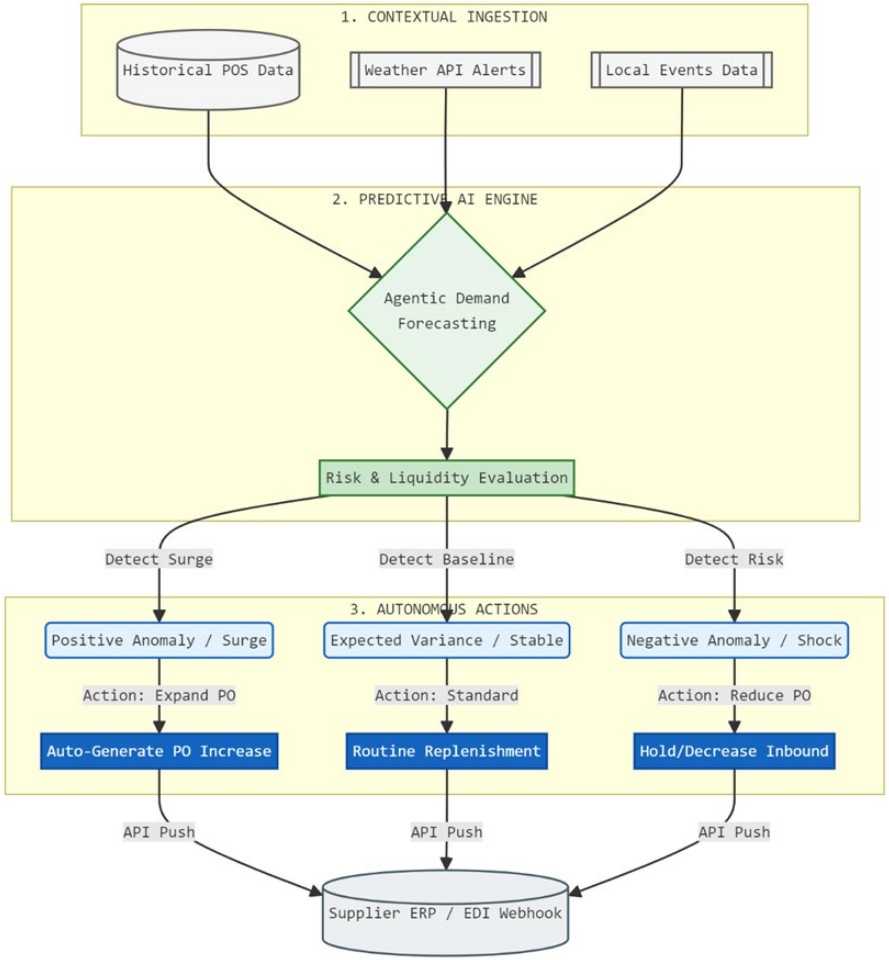


Anti-Hallucination Guardrails



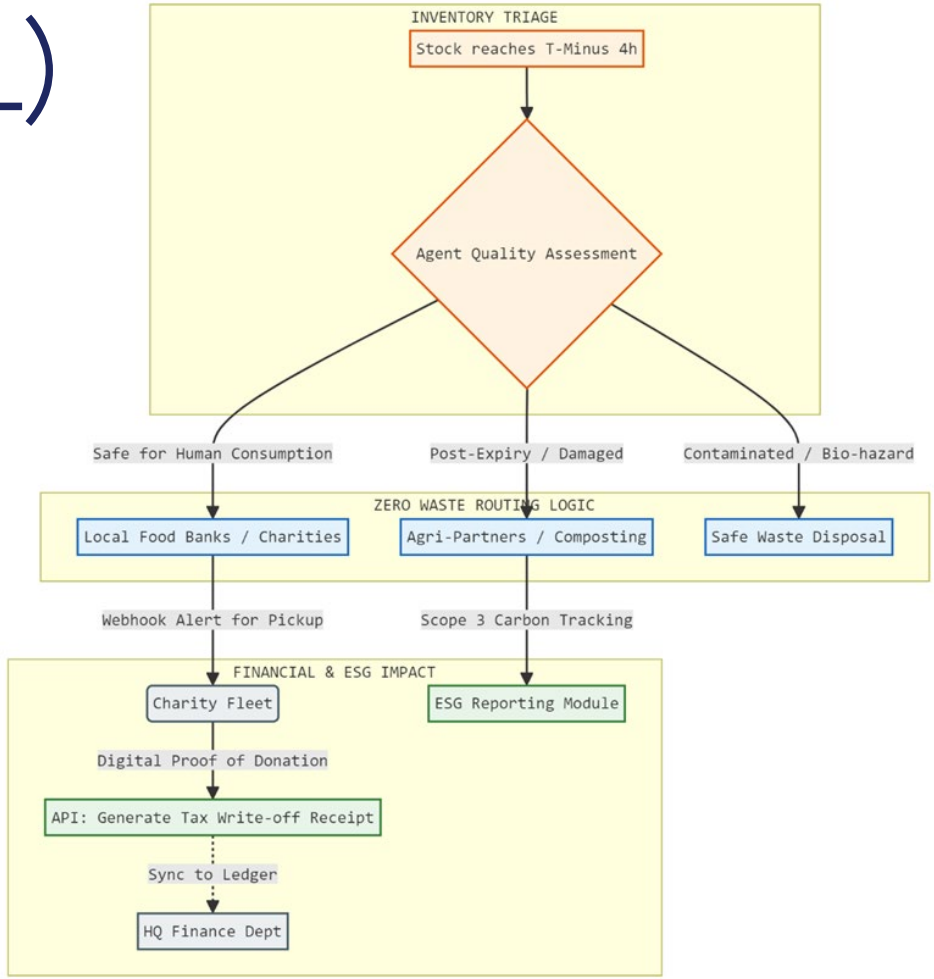
Predictive Procurement Engine

The most effective way to eliminate waste is to prevent overstocking before it enters the store. SynaptOS utilizes a predictive procurement engine that autonomously adjusts inbound Purchase Orders (POs) based on multi-variate context.



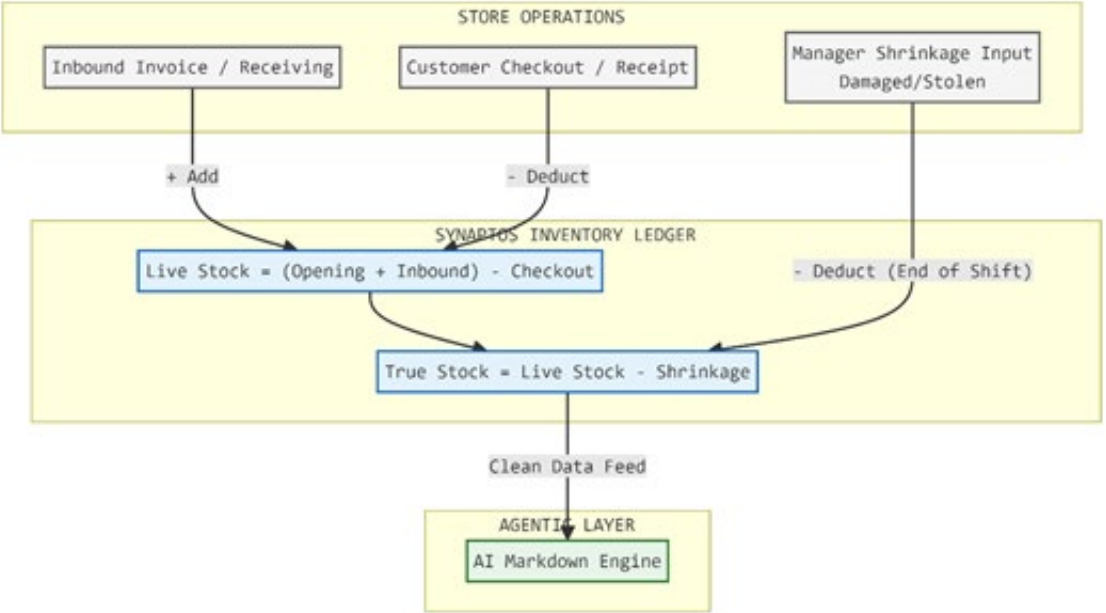
End-of-Life (EOL) waste routing

When commercial clearance (E-ink discounts) reaches its limit at T-Minus 4 hours to expire, the agent officially revokes the SKU from the sales floor and triggers physical EOL routing. This process converts unavoidable waste into ESG compliance and financial tax write-offs.



The ledger & Calibration loop

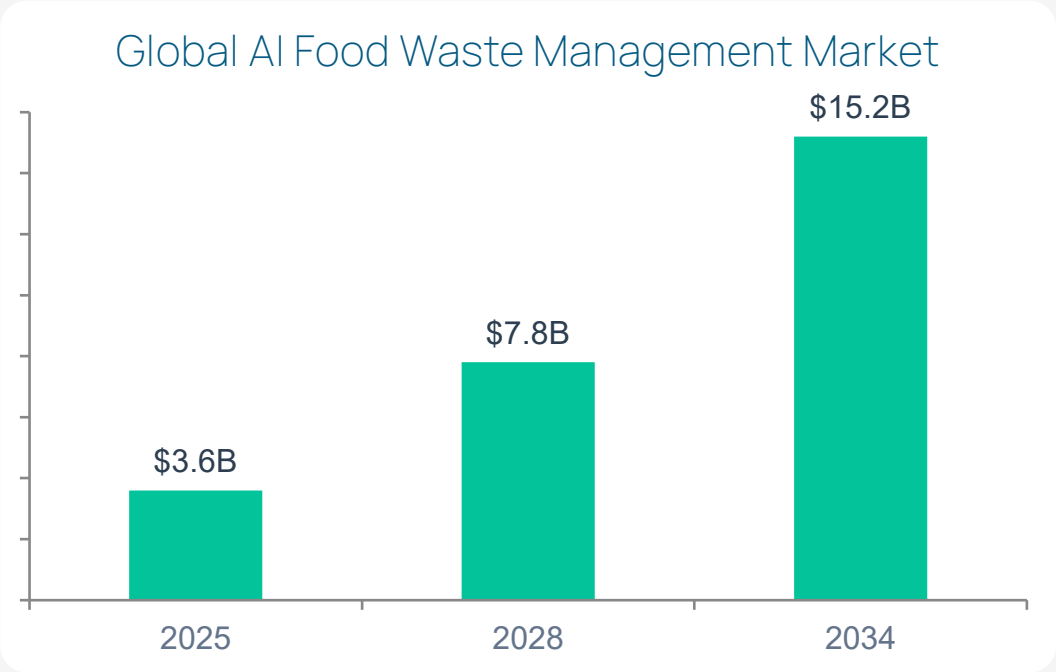
SynaptOS calculates real-time inventory purely through invoice and receipt of parsing, corrected by human intervention.



Core Capabilities

- **Contextual Dynamic Pricing**
  Progressive markdowns that adapt by district spending power, expiry window, and sale velocity
- **Virtual E-Ink Display Protocol**
  Price updates propagate to shelf labels in seconds via WebSocket — zero hardware required
- **Autonomous Procurement Agent**
  Drafts POs when stockout risk detected, preventing over-stocking that creates spoilage
- **Macro Signal Intelligence**
  Live commodity feeds enable forward-looking pricing before wholesale movements reach shelf
- **Geo-Demographic Pricing**
  District-level spending profiles set different strategies for premium vs price-sensitive zones
- **Human-in-Loop Governance**
  Manager approval for markdowns >50%, shrinkage calibration to prevent phantom inventory

Market Opportunity



MARKET SIZE

TAM
$15.2B
Global AI food waste (2034)

SAM
$850M
SEA fresh-food retail AI (2028)

SOM
$18M
Vietnam Year 4 ARR target

17.2% Market CAGR • 52.8% Retail & Supermarket Share

Why We Win?

Global players like Wasteless, RELEX, and Afresh cannot serve this market

Capability	SynaptOS	Wasteless	RELEX/Afresh
Agentic execution	✓ Autonomous	✓ Yes	✗ Dashboard only
Zero hardware	✓ Software only	✗ Requires ESL	✓ Yes
Macro signal ingestion	✓ Live commodity	✗ No	△ Partial
Geo-demographic pricing	✓ District-level	✗ No	✗ No
SEA market-native	✓ Built for Vietnam	✗ Europe/US	✗ Enterprise only
POS plugin model	✓ No rip-replace	✗ Standalone	✗ Full ERP

Defensibility: 1. Data network effects | 2. POS vendor lock-in | 3. Procurement network | 4. 18-24 month first-mover window

Business Model

SaaS Subscription

$75/store/month

Flat fee covers all AI loops, E-ink protocol, dashboard, support

Efficiency Fee

5% of rescued GMV

Performance-aligned incentive - earn more by recovering more

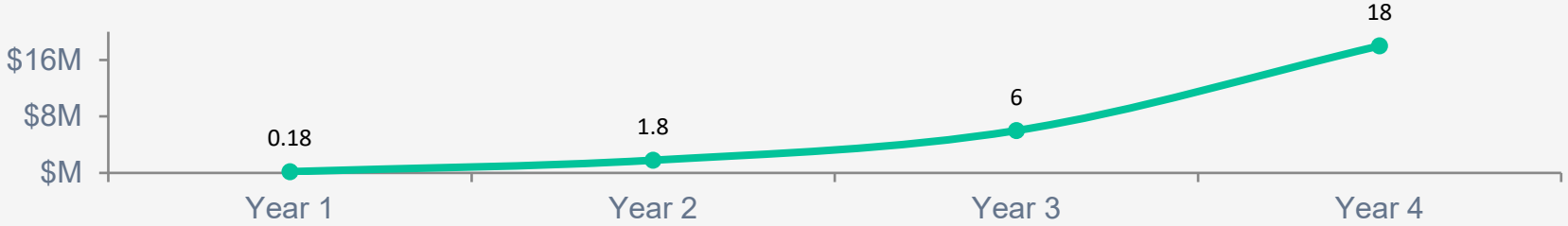
Enterprise API

$2K-15K/month

White-label pricing engine for POS vendors (Phase 2)

Go-to-Market Strategy

1. Pilot → 2. POS Channel (KiotViet / MISA) →
3. Anchor Chain → 4. Scale + SEA Expansion



Unit Economics

$0.015	$5-50	333×
Cost per AI loop	Value recovered	Min ROI

Per-Store Economics

	Per Store/Month
Total COGS	$26
SaaS subscription	$75
Efficiency fee (avg)	$40
Total revenue	$115
Gross profit	$89
Gross margin	77%

Customer Metrics

Metric	*Value*
CAC (POS channel)	$200-400
Payback period	2.3-4.5 months
Average LTV (24mo)	$2,136-2,760
LTV:CAC ratio	8-16×
Target benchmark	3×
Performance	3-5× benchmark

Marginal cost of store 1,001 is near zero - same API, same models, no incremental COGS

Execution Roadmap

Phase 1: Foundation

Months 0-6

$375K

3-5 stores, 30%+ waste reduction

- Virtual E-Ink protocol
- KiotViet integration
- Agentic loop v1
- HCMC pilot launch

Phase 2: Commercialization

Months 6-12

$525K

50 stores, $15K MRR, Seed close

- Procurement agent
- Macro signal layer
- Anchor chain deal
- SaaS billing

Phase 3: Scale

Months 12-18

$600K

200 stores, $150K MRR, Series A

- Enterprise API
- TH/ID expansion
- SDG 12 reporting
- Series A process

Every tray of chicken rescued from the bin is pure profit.

SynaptOS